

Ethics & Responsible AI — Magezi

Who Benefits

Primary Beneficiaries

- **Rural A-Level students** (est. 200,000+ in Uganda) who lack access to qualified STEM teachers, especially girls in STEM who face compounded disadvantage.
- **Teachers in understaffed schools** who can use Magezi as a teaching aid to extend their reach across subjects.
- **Parents and guardians** who want their children to succeed but cannot afford private tutoring.

Secondary Beneficiaries

- **NCDC curriculum designers** — student interaction data reveals which topics are hardest, informing curriculum revision.
- **Education researchers** studying competence-based learning in low-resource contexts.

Who Might Be Harmed

Risk: Over-reliance on AI

Mitigation: - Every response cites official NCDC syllabus pages. Students verify against the source. - Faithfulness scoring triggers “Verify with your teacher” when grounding is low. - Calibrated abstention — refuses to answer rather than guess. - Empty passage guard — Claude receives explicit “no context” instead of hallucinating. - Agentic tutor planner decides complexity and approach BEFORE generating. - Feedback buttons (thumbs up/down) on every response via TanStack mutation.

Risk: Exam Cheating

Mitigation: - Input guardrails detect and block “give me exam answers” and UNEB answer requests. - Pedagogical pipeline (explain → example → activity → try → feedback) builds understanding. - System prompt refuses “just give the answer” without pedagogical scaffolding.

Risk: Cultural & Linguistic Harm

Mitigation: - Code-switching preserves English scientific terms — never force-translates “photosynthesis” or “acceleration”. - 120-term multilingual bridge (Luganda, Swahili, Runyankole) enables cross-language retrieval without mistranslation. - Feedback buttons allow students to report language errors.

Risk: Data Privacy

Mitigation: - **Uganda NDPA SS19 compliance:** Sessions auto-expire (24h TTL, configurable). - No PII collected — no names, school IDs, or location data required. - Thread-safe session store bounded to 5K sessions max. - Auth passwords bcrypt-hashed. JWT tokens expire in 72h. - Feedback store bounded to 10K entries (no unbounded growth).

Risk: Digital Divide Amplification

Mitigation: - PWA with service worker — UI works offline after initial load. - Voice-first design — Web Speech API + TTS for low-end phones. - Mobile-first layout — 100dvh chat, sticky composer, collapsible hero. - Offline detection banner shows “You are offline” with graceful degradation.

Safeguards

Safeguard	Implementation
Input filtering	OWASP LLM01 prompt-injection regex + education content + exam cheating block
LLM resilience	Groq retry (3x backoff on 429) → local fallback — student always gets an answer
Agentic planning	Tutor analyses complexity, decides multi-retrieval and tool use BEFORE answering
Output grounding	Faithfulness score with threshold — low scores trigger “verify” banner
Calibrated abstention	Refuses when retrieval confidence is too low
Empty passage guard	Claude receives “no context” instead of hallucinating from nothing
	18 constants + 47 formulas injected for numeric accuracy

Safeguard	Implementation
STEM tool verification	
Content guardrails	Refuses harmful, illegal, age-inappropriate, or off-topic content
Escalation	Routes non-STEM queries to “ask your teacher”
Rate limiting	Thread-safe sliding window (30 req/60s), bounded memory
Session TTL	24h auto-expiry, 5K session max
Auth security	bcrypt passwords, JWT (72h expiry), BYOK key storage
CORS hardening	Configurable origins, credentials disabled
Feedback loop	Thumbs up/down on every response

Alignment with Hackathon Values

AI That Empowers Humans

Magezi is a **supplement, not a replacement** for teachers. It follows the exact NCDC pedagogy teachers are trained in. Every response encourages verification with the teacher.

Solving a Real Problem

The teacher-to-student ratio in rural Ugandan A-Level schools averages 1:80+. The 2025 competence-based curriculum demands interactive, activity-based learning that a single teacher cannot deliver to 80 students. Magezi fills the gap between lessons.

Economic Empowerment Through Education

A-Level STEM qualifications unlock university admission in engineering, medicine, agriculture, and technology — careers that drive Uganda’s economic transformation.

Regulatory Compliance

Regulation	Status
Uganda National Data Protection Act (NDPA) 2019	Compliant — SS19 data minimisation, session TTL, bounded stores

Regulation	Status
NCDC Curriculum Guidelines 2025	Aligned — uses official 288-page syllabi as knowledge base
UNEB Examination Ethics	Respects — teaches methodology, blocks answer-giving
WCAG 2.2 AA Accessibility	Compliant — skip-link, focus-visible, reduced-motion, contrast 4.7:1

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